

PHANTOM TEMPO™

Framework for Detecting Machine-Speed Adversary Operations and Agentic Integrity Failures

Version 0.2 - Public Draft

THE SPEED BECOMES THE INDICATOR OF ATTACK

A vendor-neutral framework for measuring velocity, parallelism, adaptation, coordination, and decision-cycle compression in adversary behavior.

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Status and Intended Use

This document defines PHANTOM TEMPO™, a public, vendor-neutral defensive framework for identifying adversary operations whose timing, adaptive behavior, state integrity, and objective stability indicate AI assistance, autonomous agents, high-order orchestration, or other machine-speed execution.

Version 0.2 is an expanded public draft specification. It is intended for security vendors, defenders, researchers, red teams, incident responders, data scientists, standards bodies, and operators of identity, endpoint, cloud, network, application, database, and security analytics platforms.

The framework does not claim that speed alone proves artificial intelligence. It provides a structured method to identify and explain when velocity, parallelism, adaptation, coordination, and decision-cycle compression collectively exceed appropriate human and approved-automation baselines.

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Normative Language

The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, MAY, and OPTIONAL are to be interpreted as described by BCP 14 (RFC 2119 and RFC 8174) only when they appear in all capitals. Because this is a pre-standard public draft, normative requirements remain subject to review and validation.

Framework Principles

- Behavior over attribution. PHANTOM TEMPO evaluates observable operations, not unverifiable claims about the actor or model.
- Tempo over volume. High event counts are insufficient without temporal structure, adaptation, or coordination.
- Cross-domain correlation. Meaningful tempo often spans identities, hosts, applications, cloud services, and data stores.
- Explainability. Every score or alert must identify the observations that produced it.
- Vendor neutrality. The framework defines outcomes and measurements, not a required product architecture.
- Open implementation. Defensive products may map to and implement the framework under the published brand and use policy.
- Telemetry-grounded integrity. Hallucination and drift are assessed through observable contradictions and behavioral divergence, not claims about an AI system's internal reasoning.
- Separate conclusions. Tempo, agentic integrity, objective stability, maliciousness, and model attribution are distinct judgments and MUST NOT be collapsed into one score.

1. Purpose

Traditional security detections focus on known artifacts, known procedures, threshold violations, or isolated behavioral anomalies. AI-assisted and autonomous adversaries can execute familiar techniques while compressing the time between discovery, decision, correction, and exploitation. They can also explore several paths simultaneously and coordinate activity across technologies faster than a human operator can plausibly reason, type, validate, and redirect.

PHANTOM TEMPO adds a temporal and adaptive analysis layer to existing security models. It is designed to answer a specific defensive question:

DOES THE CADENCE, CONCURRENCY, ADAPTATION, AND COORDINATION OF THIS ACTIVITY INDICATE MACHINE-SPEED ADVERSARY OPERATION?

The framework also evaluates operational integrity failures associated with agentic systems, including unsupported success claims, state hallucination, context loss, corrective oscillation, and drift from an inferred objective or constraint set. These signals can reveal agentic operation even when an adversary deliberately slows its activity to resemble human behavior.

2. Scope

PHANTOM TEMPO applies to security-relevant activity involving:

- Identity and access systems
- Endpoints, servers, containers, and workloads
- Cloud control planes and SaaS platforms
- Network, DNS, proxy, and remote access infrastructure
- Applications, APIs, databases, and data platforms
- Email, collaboration, and messaging systems
- Operational technology and connected devices
- Security controls, administrative tools, and defensive infrastructure

The framework may be used for real-time detection, retrospective hunting, incident reconstruction, threat research, red-team testing, product evaluation, control validation, and security architecture.

3. Non-Goals

- Proving that a specific commercial or open-source AI model was used.
- Replacing MITRE ATT&CK®, MITRE D3FEND™, OCSF, or other existing taxonomies and schemas.
- Classifying every fast script, scanner, bot, or administrator as malicious.
- Defining offensive procedures or providing operational guidance for intrusion.
- Producing a single universal threshold that is valid for every environment.
- Inferring private chain-of-thought or diagnosing a model’s internal cognitive state. PHANTOM TEMPO evaluates externally observable claims, actions, results, and contradictions.

4. Core Definitions

Term	Definition
Adversary Action	A security-relevant operation performed or initiated by an actor, tool, script, service, or autonomous agent.

Term	Definition
Action Rate	The number of relevant actions within a defined interval, adjusted for action type and target context.
Concurrency	The degree to which distinct action paths overlap in time.
Adaptation	A change in behavior that is causally responsive to a result, error, state change, or newly discovered condition.
Adaptation Latency	The elapsed time between an observed result and the first behavior that meaningfully responds to it.
Coordination	The purposeful alignment of related activity across identities, systems, services, or attack paths.
Decision-Cycle Compression	Reduction in the time required to observe, interpret, choose, execute, and validate an action.
Machine-Speed Adversary Operation	Activity whose combined temporal and behavioral properties are inconsistent with ordinary human operation and are consistent with automation, AI assistance, autonomous agents, or orchestrated machine execution.
Tempo Window	The bounded period over which related behavior is measured and correlated.
Phantom Tempo Event	A correlated set of observations that satisfies one or more PHANTOM TEMPO behavioral patterns.
Phantom Tempo Score	An explainable composite estimate of machine-speed characteristics, not a declaration of AI attribution.
Persistence Modifier	A measure of how long or how often elevated tempo recurs. Persistence strengthens evidence but is not one of the five core dimensions.
Agentic Integrity	The degree to which an operation's assertions, retained state, dependent actions, and reported outcomes remain consistent with authoritative telemetry.

Term	Definition
Operational Hallucination	An externally observable assertion or dependent action that is unsupported or contradicted by authoritative telemetry. This term describes operational residue, not private model reasoning.
State-Telemetry Divergence	A material mismatch between the state represented by an actor or agent and the state recorded by authoritative systems.
Objective Stability	The degree to which activity remains aligned with an inferred operational objective and previously established constraints.
Goal Drift	A measurable divergence from the inferred operational objective that cannot be explained by new evidence, a legitimate phase transition, or explicit operator direction.
Constraint Drift	Progressive or abrupt violation of previously observed scope, policy, safety, or operating constraints.
Authoritative Telemetry	The source or set of sources designated as the best available evidence for whether a claimed state transition or result occurred.

5. The Five Tempo Dimensions

PHANTOM TEMPO measures five dimensions. Persistence is treated as an evidence-strength modifier rather than a sixth dimension. No single dimension is sufficient in all cases; confidence increases when multiple dimensions reinforce one another.

Dimension	Meaning	Representative Measurements
PT-D1: Velocity	How quickly actions occur relative to environmental and human baselines.	Inter-action interval; actions per second/minute; sequence duration; target acquisition rate.
PT-D2: Parallelism	How many independent or partially independent paths are pursued concurrently.	Concurrent branches; target fan-out; overlapping sessions; simultaneous protocol or credential tests.

Dimension	Meaning	Representative Measurements
PT-D3: Adaptation	How rapidly and specifically behavior changes in response to observed outcomes.	Failure-to-correction latency; parser or command substitution; hypothesis testing; successful corrective retry.
PT-D4: Coordination	How tightly related actions align across identities, systems, or control planes.	Cross-domain sequence continuity; synchronized changes; shared objectives; credential-to-resource pivots.
PT-D5: Decision-Cycle Compression	How much the complete observe-orient-decide-act-validate sequence is compressed relative to an appropriate baseline.	End-to-end loop duration; stage-transition latency; decision-to-action latency; baseline compression ratio.

6. Analytical Model

A PHANTOM TEMPO analysis begins by constructing an ordered activity graph. Nodes represent actions, results, identities, targets, and artifacts. Directed edges represent temporal order, causal response, shared context, or inferred operational dependency.

A conforming implementation SHOULD calculate at least the following measurements:

Code	Measurement	Definition
IAT	Inter-Action Time	Elapsed time between related actions.
FCR	Failure-to-Correction Response	Elapsed time from a failed action to a causally related corrective action.
BPF	Branching Path Factor	Number of distinct investigative or execution paths opened within a tempo window.
COI	Concurrency Overlap Index	Proportion of the tempo window containing two or more active paths.
CTC	Cross-Technology Coordination	Strength of correlated activity across technical domains.

Code	Measurement	Definition
TAR	Target Acquisition Rate	Rate at which new relevant resources, accounts, hosts, or services are discovered and acted upon.
VLR	Validation Loop Rate	Frequency of execute-observe-adjust cycles.
DCR	Decision-Cycle Compression Ratio	Baseline decision-cycle duration divided by the observed duration for a comparable sequence.
PES	Persistence of Elevated Speed	Duration or recurrence of elevated tempo; used as an evidence-strength modifier rather than a core dimension.

7. Baselines and Context

Tempo is contextual. Products **MUST** avoid universal assumptions that a specific number of actions per second is malicious. Baselines **SHOULD** be segmented by identity type, asset type, tool, workflow, administrative function, time of day, and expected automation.

- Human interactive baseline: ordinary operator workflows, including experienced administrators and responders.
- Approved automation baseline: orchestration, deployment, scanning, backup, monitoring, and service accounts.
- Environmental burst baseline: patch windows, failover events, incident response, and bulk migrations.
- Peer baseline: comparable identities, hosts, applications, and workloads.
- Historical self-baseline: the same actor, system, or service over time.

A detection **MUST** support allowlisting or contextual suppression for approved automation, but **SHOULD** still retain telemetry so that compromised automation can be evaluated for changed purpose, target set, or adaptive behavior.

8. PHANTOM TEMPO Behavioral Pattern Catalog

Pattern identifiers are stable labels for observable machine-tempo behavior. They do not replace attack techniques. A single offensive technique may exhibit several tempo patterns, and one tempo pattern may span many offensive techniques.

ID	Pattern	Definition	Required Evidence	Primary Measures
PT-B1001	Accelerated Failure Correction	A failed action is followed by a technically relevant corrective action within an interval inconsistent with the normal workflow.	Failure event; diagnostic probe; changed command, credential format, parser, protocol, or parameter; validation result.	Failure-to-correction latency; specificity of correction; success after correction.
PT-B1002	Parallel Hypothesis Testing	Multiple competing explanations or paths are tested concurrently or in rapid interleaving rather than serially.	Distinct queries, authentication modes, protocols, endpoints, or privilege paths launched within the same tempo window.	Branching path factor; concurrency overlap; path diversity.
PT-B1003	Rapid Target Fan-Out	The operation discovers and acts against a growing set of targets faster than ordinary human selection and execution.	Host, account, resource, database, API, mailbox, or secret enumeration followed by targeted actions.	Target acquisition rate; action-to-target ratio; fan-out acceleration.
PT-B1004	Cross-Domain Operational Continuity	The same objective continues coherently across multiple technology domains without ordinary handoff delay.	Identity event followed by cloud, endpoint, application, database, or network action sharing context.	Cross-technology coordination; transition latency; objective continuity.
PT-B1005	Compressed Execute-Validate-Adjust Loop	The actor repeatedly executes an action, evaluates its result, changes course, and validates the new action at machine tempo.	Alternating execution and observation events with result-dependent modifications.	Validation loop rate; median adaptation latency; successful adjustment ratio.

ID	Pattern	Definition	Required Evidence	Primary Measures
PT-B1006	Adaptive Tool or Parser Substitution	A tool, parser, payload format, or method is replaced in direct response to incompatible output or failure.	Error or malformed output followed by a fit-for-purpose replacement.	Substitution latency; semantic relevance; subsequent success.
PT-B1007	Credential-Resource Pivot Compression	Newly discovered credentials, tokens, keys, or secrets are tested against relevant resources with minimal delay.	Secret discovery followed by authentication attempts or API use across likely targets.	Discovery-to-use latency; number of resource classes tested; credential fan-out.
PT-B1008	Coordinated Multi-Identity Operation	Multiple identities or sessions pursue one objective with synchronized timing or shared state.	Overlapping account, token, workload identity, or session activity converging on related resources.	Identity concurrency; shared target overlap; synchronized transitions.
PT-B1009	Defensive Feedback Evasion	Behavior changes quickly after blocks, detections, throttling, or policy enforcement, indicating feedback-driven evasion.	Control alert or denial followed by altered timing, target, protocol, identity, or payload characteristics.	Block-to-evasion latency; evasion diversity; recurrence.
PT-B1010	Sustained High-Order Orchestration	Machine-speed behavior remains coherent across several stages rather than appearing as a single scripted burst.	Persistent multi-stage sequence with repeated adaptation and coordination.	Persistence of elevated speed; stage count; cross-stage context retention.

8.1 Pattern-to-Dimension Mapping

This mapping makes conformance claims testable. Primary dimensions are expected to carry the strongest evidence; secondary dimensions may contribute when telemetry supports them.

Pattern	Primary Dimension(s)	Secondary Dimension(s)	Persistence Use
PT-B1001	Adaptation; Decision-Cycle Compression	Velocity	Recurrence raises confidence
PT-B1002	Parallelism	Velocity; Coordination	Sustained branching raises confidence
PT-B1003	Velocity; Parallelism	Coordination	Sustained fan-out raises confidence
PT-B1004	Coordination; Decision-Cycle Compression	Velocity	Cross-stage continuity raises confidence
PT-B1005	Adaptation; Decision-Cycle Compression	Velocity	Repeated loops raise confidence
PT-B1006	Adaptation	Decision-Cycle Compression	Repeated substitutions raise confidence
PT-B1007	Decision-Cycle Compression; Coordination	Velocity; Parallelism	Repeated pivots raise confidence
PT-B1008	Coordination; Parallelism	Velocity	Sustained identity coordination raises confidence
PT-B1009	Adaptation; Decision-Cycle Compression	Coordination	Repeated control-response changes raise confidence
PT-B1010	Coordination; Decision-Cycle Compression	All other dimensions	Persistence is intrinsic to this pattern

8.2 Agentic Integrity and Objective Stability

PHANTOM TEMPO treats tempo and agentic integrity as complementary analytical pillars. Tempo measures how an operation moves. Agentic integrity measures whether its asserted state and dependent behavior remain grounded in observed reality. Objective stability measures whether the operation remains aligned to its inferred goal and constraints. An implementation **MUST** report these conclusions separately.

Operational hallucination is detected through contradictions such as a credential recorded as valid without a successful authentication event, continuation into post-exploitation without a confirmed session, or classification of public information as newly obtained sensitive material. Drift is detected through unexplained changes in target, scope, priority, or constraints. Neither condition alone proves that an LLM was used.

8.2.1 Agentic Integrity Pattern Catalog

ID	Pattern	Definition	Required Evidence	Primary Measures
PT-I1001	Unsupported Credential Claim	A credential, token, key, or session is represented as valid without corroborating successful use.	Claimed-valid artifact; authoritative authentication result; dependent action.	Claim-to-validation gap; failed-use count; downstream dependency.
PT-I1002	Fabricated Success Continuation	The operation advances to a stage that requires a prerequisite success that telemetry does not confirm.	Prerequisite action and result; subsequent dependent action; absence of required state transition.	Dependency violation count; continuation latency; affected stages.
PT-I1003	Unsupported Intelligence Classification	Information is classified as secret, sensitive, proprietary, or newly acquired without supporting provenance or despite evidence that it is public.	Agent classification; provenance checks; data source and access history.	Classification contradiction; provenance confidence; repeated overstatement.
PT-I1004	State-Telemetry Divergence	The operation's represented state differs materially from authoritative identity, endpoint, cloud, application, network, or data telemetry.	Represented state; authoritative event sequence; time-bounded comparison.	Divergence duration; severity; number of dependent actions.
PT-I1005	Hallucination-Correction Loop	An unsupported assumption generates failures, retries, or corrective actions before the operation returns to a telemetry-supported state.	Initial unsupported assumption; resulting actions; failures; correction or abandonment.	Loop duration; wasted branches; correction latency.

ID	Pattern	Definition	Required Evidence	Primary Measures
PT-I1006	Goal Drift	Activity diverges from the inferred objective without a telemetry-supported reason, explicit instruction, or legitimate phase transition.	Prior objective evidence; changed target or task; absence of justified transition.	Objective distance; drift duration; scope expansion.
PT-I1007	Corrective Oscillation	The operation alternates repeatedly among incompatible approaches without accumulating stable state or resolving the triggering condition.	Alternating actions or hypotheses; repeated failures; limited net progress.	Oscillation count; cycle period; state reset frequency.
PT-I1008	Cross-Stage Context Loss	Confirmed facts, exclusions, credentials, targets, or constraints are not retained across operational stages or agent handoffs.	Earlier confirmed state; later contradictory or redundant action; stage/session relationship.	Context-loss recurrence; redundant work; contradiction severity.
PT-I1009	Invalid Capability Assumption	The operation attempts an action that the current identity, tool, protocol, host, or resource cannot perform and proceeds as though the capability exists.	Capability requirements; actual privileges or interfaces; attempted and dependent actions.	Capability mismatch; continuation count; correction latency.

ID	Pattern	Definition	Required Evidence	Primary Measures
PT-I1010	Constraint or Policy Drift	The operation progressively or abruptly violates previously established scope, safety, targeting, or execution constraints.	Established constraints; violating actions; timing and escalation pattern.	Constraint violations; drift slope; affected control domains.

8.2.2 Three-Axis Analytical Model

A conforming implementation SHOULD report three independent axes: (1) Tempo Signature - velocity, parallelism, adaptation, coordination, and decision-cycle compression; (2) Agentic Integrity Divergence - unsupported state, contradiction, fabricated success, and context failure; and (3) Objective Stability - goal and constraint alignment over time. Maliciousness and model attribution remain separate determinations.

Conclusion	Question Answered	Evidence Basis
Tempo Signature	How strongly the action graph exhibits machine-speed characteristics.	PT-D dimensions and PT-B patterns.
Agentic Integrity Divergence	How strongly represented state and dependent behavior conflict with authoritative telemetry.	PT-I1001 through PT-I1005, PT-I1007 through PT-I1009.
Objective Stability	How consistently the activity remains aligned to inferred goals and constraints.	PT-I1006 and PT-I1010, with supporting sequence evidence.
Maliciousness	Whether the activity is unauthorized or harmful.	Independent security context and incident evidence.
Model Attribution	Whether a particular AI system or model family was used.	Provider, forensic, infrastructure, or actor evidence; never inferred from PT scoring alone.

9. Detection Confidence

PHANTOM TEMPO confidence expresses the strength of evidence for a machine-tempo signature. It does not independently determine malicious intent, compromise, or use of an LLM. Those determinations belong in a separate incident disposition supported by additional evidence.

Confidence	Interpretation
PT-C0 — Not Assessed	Insufficient telemetry, timing quality, or correlation.
PT-C1 — Anomalous Tempo	One or more dimensions exceed an appropriate baseline, but ordinary automation or burst activity remains plausible.
PT-C2 — Coordinated Machine Execution	Tempo combines with parallelism or cross-domain coordination in a way consistent with machine execution.
PT-C3 — Adaptive Machine Execution	Result-dependent adaptation is observed at abnormal latency with a defensible causal relationship.
PT-C4 — Sustained Adaptive Orchestration	Multiple adaptive loops, parallel exploration, coordination, and compressed decision cycles form a coherent sustained operation.

10. Reference Scoring Model

The reference score is experimental and non-normative. It is included to make the draft testable, not to establish validated universal weights or thresholds. Vendors may use other models, but outputs must remain explainable and map to the five tempo dimensions.

Normalize each dimension to a value from 0 to 100 against the appropriate baseline:

$$\text{PTS}(\text{exp}) = 0.20V + 0.15P + 0.30A + 0.15C + 0.20D$$

Where V = velocity, P = parallelism, A = adaptation, C = coordination, and D = decision-cycle compression. Adaptation receives the highest provisional weight because result-dependent correction is generally more discriminating than raw speed. These weights have not yet been empirically validated; implementations must disclose deviations and must not compare scores across products as though they were equivalent.

Score	Illustrative Evidence Band	Recommended Handling
0–24	Expected / baseline-consistent	Retain for baseline; no PT alert.
25–44	Elevated tempo evidence	Enrich and correlate.
45–64	Suspicious machine-tempo signature	Create analytic event; evaluate approved automation.

Score	Illustrative Evidence Band	Recommended Handling
65–79	Strong adaptive-execution evidence	Escalate with contributing patterns and cross-domain evidence.
80–100	High-confidence sustained adaptive orchestration	Prioritize incident response and containment review.

Any score must include contributing measurements, baseline selection, confidence limitations, model/version information, and the specific patterns observed. Products must not expose only an opaque numeric value. The score bands below are illustrative assessment aids and are not certification thresholds.

10.1 Agentic Integrity and Objective-Stability Reporting

The experimental PTS score applies only to tempo. Implementations **MUST NOT** add integrity failures or drift directly into PTS. They **SHOULD** emit separate ordinal assessments so a fast but coherent script, a slow agent with hallucination residue, and a fast unstable agent remain distinguishable.

Level	Interpretation	Minimum Characterization
AID-0 / OS-0	No assessment	Insufficient state, provenance, or objective context.
AID-1 / OS-1	Weak anomaly	Single contradiction or possible objective change with plausible benign explanation.
AID-2 / OS-2	Correlated divergence	Multiple related contradictions, dependent actions, or unexplained scope changes.
AID-3 / OS-3	Strong agentic-integrity or drift signature	Repeated state-telemetry divergence, fabricated continuation, context loss, or sustained objective/constraint drift.
AID-4 / OS-4	High-confidence operational instability	Coherent multi-stage evidence of hallucination residue or drift with material operational impact.

Example report: “Tempo PT-C3; Agentic Integrity AID-3; Objective Stability OS-1; maliciousness confirmed independently; model attribution unconfirmed.”

11. Minimum Telemetry Requirements

Domain	Minimum Fields
Common	High-resolution timestamp; source; event type; actor/identity; target; result; session or trace identifier; action identifier.
Identity	Authentication method; token or credential type; application; device; IP; conditional access result; failure reason.
Endpoint/Workload	Process lineage; command or action metadata; file/network relationships; execution result; container or workload identity.
Network	Source/destination; protocol; session timing; DNS; proxy; connection outcome; byte and packet context where available.
Cloud/SaaS	API operation; resource; principal; region; request ID; response code; policy or control result.
Application/API	Route or method; caller; input class; response status; trace ID; latency; parser or validation errors.
Database/Data	Principal; source; query or operation class; schema/object; result; row or object count; transaction context.
Security Controls	Alert, block, throttle, challenge, quarantine, or policy action with timestamp and target.
Agent/Orchestrator State	Task identifier; parent/sub-agent relationship; asserted result; retained facts; objective or plan version; tool call and response correlation; human approval or override.
Provenance and Validation	Claimed source; public/private classification basis; credential validation status; prerequisite-success evidence; authoritative source designation.
Policy and Scope	Declared targets; allowed actions; safety or operational constraints; changes to scope; approver and reason.

Millisecond timestamps are RECOMMENDED. Microsecond or nanosecond precision MAY be retained when natively available. Products MUST preserve ordering and SHOULD preserve clock quality or uncertainty metadata.

12. Correlation Requirements

- Correlate by more than IP address. Use identity, session, token, process, request, trace, host, workload, and target relationships.
- Support many-to-many relationships between actions and targets.
- Preserve failed actions, denials, malformed responses, and control feedback; these are essential to adaptation analysis.
- Distinguish repeated identical actions from result-dependent changes.
- Support overlapping windows and multi-stage incidents.
- Represent uncertainty when causal relationships are inferred rather than directly observed.
- Correlate asserted or recorded agent state with authoritative telemetry and preserve the source of each assertion.
- Represent prerequisite relationships so dependent actions can be evaluated against confirmed success or failure.
- Preserve objective, scope, constraint, and plan changes with the actor or process responsible for each change.
- Distinguish explicit operator redirection from unexplained goal drift or context loss.

13. Alert and Record Schema

A PHANTOM TEMPO analytic record SHOULD contain:

Field	Description
pt_event_id	Unique identifier.
framework_version	PHANTOM TEMPO version used.
start_time / end_time	Tempo window boundaries.
actors	Related users, service accounts, workload identities, sessions, or processes.
targets	Related systems, resources, services, identities, and data objects.
patterns	Observed PT-B identifiers.
dimensions	Normalized dimension values and raw measurements.
score	Composite score, if used.

Field	Description
confidence	PT-C level.
baseline_context	Baseline cohort and comparison period.
evidence	Ordered observations and causal or temporal relationships.
approved_automation_status	Known, unknown, changed, or not applicable.
attack_mappings	Optional MITRE ATT&CK technique mappings.
defensive_mappings	Optional MITRE D3FEND countermeasure mappings.
disposition	Open, benign automation, compromised automation, confirmed incident, or inconclusive.
integrity_patterns	Observed PT-I identifiers.
agent_assertions	Claims or retained state attributed to an agent, tool, or orchestrator.
authoritative_evidence	Telemetry used to confirm or contradict each material assertion.
integrity_level	AID-0 through AID-4 with explanation.
objective_state	Inferred objective, confidence, and material changes over time.
objective_stability_level	OS-0 through OS-4 with explanation.
operator_interventions	Human approvals, corrections, overrides, or phase transitions.

14. Relationship to Existing Frameworks

PHANTOM TEMPO is an analytical overlay. MITRE ATT&CK describes how adversaries interact with systems through tactics and techniques based on real-world observations. MITRE D3FEND provides a knowledge graph of defensive countermeasure techniques. OCSF provides an implementation-agnostic schema for cybersecurity events. PHANTOM TEMPO adds a normalized way to measure the velocity, parallelism, adaptation, coordination, and decision-cycle compression with which those actions occur.

Resource	PHANTOM TEMPO Use
MITRE ATT&CK	Map PT events to the offensive techniques present in the underlying activity.
MITRE D3FEND	Map recommended countermeasures and defensive capabilities to PT patterns.
OCSF	Represent source events and, where extended, PT analytic records in a vendor-neutral schema.
Detection engineering formats	Express product-specific rules, correlation logic, thresholds, and hunting content.
Incident response frameworks	Use PT confidence and evidence to prioritize containment and investigative sequencing.

15. Public Case Mappings

15.1 GTG-1002 Agentic Integrity Case Mapping

Anthropic disclosed GTG-1002 on November 13, 2025 and assessed that a Chinese state-sponsored group used Claude Code in a largely autonomous espionage campaign conducted in September 2025. Anthropic reported that Claude frequently overstated findings and occasionally fabricated data, including claims of credentials that did not work and “critical discoveries” that were publicly available. These observations provide direct public evidence of operational hallucination residue in a live cyber campaign. Anthropic did not publicly document clear goal drift, so GTG-1002 MUST NOT be cited as confirmed goal-drift evidence.

Reported Observation	PT Mapping	Defensive Interpretation
Claimed credentials that failed validation	PT-I1001, PT-I1004	Credential claim versus authentication telemetry; dependent actions.
Critical discoveries later found to be public	PT-I1003, PT-I1004	Classification and provenance contradiction.
Need for operator validation of claimed results	PT-I1004, PT-I1005	Frequency and consequence of unsupported claims; correction loops.
No documented mission divergence	No confirmed PT-I1006 mapping	Insufficient public evidence of goal drift; do not overclaim.

15.2 JADEPUFFER Initial Case Mapping

On July 1, 2026, the Sysdig Threat Research Team reported what it assessed as an end-to-end LLM-driven database extortion operation named JADEPUFFER. Its report describes purposeful payload generation, parallel investigation, rapid error correction, credential use, and continuation from an exposed Langflow instance to a production database objective. This framework treats those claims as attributed research findings, not as independently verified telemetry.

Case Observation	PT Mapping	Key Measurement
Observed rapid correction after authentication failure	PT-B1001, PT-B1005	Failure-to-correction response; validation loop rate.
Competing explanations tested in close succession	PT-B1002	Branching path factor; concurrency overlap.
Change in processing after unexpected output	PT-B1006	Substitution latency; subsequent success.
Movement from initial service compromise toward production database objective	PT-B1004, PT-B1007	Cross-domain transition latency; credential discovery-to-use.
End-to-end coherent extortion activity	PT-B1010	Persistence of elevated speed; cross-stage continuity.

The JADEPUFFER mapping is illustrative, not a complete independent reconstruction. PHANTOM TEMPO should be applied to original telemetry whenever available.

No single product is expected to collect every telemetry domain or perform every capability. Implementations may be distributed across telemetry producers, correlators, analytics platforms, controls, and responders, provided the combined system preserves evidence and exposes the required mappings.

16. Product Implementation Profiles

Profile	Required Capability
PT-Observe	Collects and preserves telemetry needed for tempo analysis with adequate timestamp quality and identity/target context.
PT-Correlate	Builds cross-event and cross-domain relationships, including failures, results, and causal responses.
PT-Detect	Implements one or more PT behavioral patterns and emits explainable events.

Profile	Required Capability
PT-Score	Calculates dimension values, confidence, and optionally a composite score against contextual baselines.
PT-Respond	Uses PT events to drive enrichment, containment recommendations, or automated response with safeguards.
PT-Validate	Supports assessment datasets, emulation, repeatable tests, and evidence export for conformance evaluation.
PT-Integrity	Correlates assertions and dependent actions with authoritative telemetry and detects PT-I integrity patterns.
PT-Objective	Maintains objective and constraint state, identifies justified transitions, and detects unexplained drift.

17. Conformance Levels

Level	Criteria
Level 1 — Mapped	Product documentation maps existing capabilities to PT dimensions or patterns. No conformance claim beyond mapping.
Level 2 — Implemented	Product natively detects at least three patterns across at least two dimensions and provides explainable evidence.
Level 3 — Cross-Domain	Product correlates at least three telemetry domains, measures adaptation plus coordination, and implements at least two PT-I patterns using authoritative-state comparison.
Level 4 — Adaptive Defense	Product supports sustained cross-domain tempo detection, agentic-integrity and objective-stability reporting, validated testing artifacts, explainable evidence, and response integration.

Until a formal validation program exists, vendors should use factual language such as “implements mappings to the PHANTOM TEMPO™ Framework v0.2” rather than “certified,” “validated,” or “conformant.”

18. Assessment Methodology

A defensible product assessment SHOULD test the following:

1. Timestamp fidelity and ordering under realistic event volume.
2. Recognition of approved fixed automation without unnecessary escalation.
3. Detection of a fast but non-adaptive script.
4. Detection of result-dependent correction after a controlled failure.
5. Detection of two or more competing paths pursued in parallel.
6. Correlation of one objective across identity, endpoint/workload, application, cloud, or database telemetry.
7. Explainable output containing raw evidence and calculated measurements.
8. Resilience to missing, delayed, duplicated, or out-of-order telemetry.
9. Identification of compromised approved automation whose targets or behavior changed.
10. Repeatability of results across multiple runs and environments.
11. Detection of a claimed-valid credential contradicted by authoritative authentication telemetry.
12. Detection of dependent-stage continuation after a controlled prerequisite failure.
13. Detection of unsupported sensitivity classification using known-public test material.
14. Detection of cross-stage context loss and corrective oscillation.
15. Distinction between explicit operator redirection and unexplained goal drift.
16. Detection of constraint drift while preserving legitimate scope-change records.

19. False Positives and Confounders

Confounder	Disambiguation Guidance
Approved orchestration	May be fast and parallel, but generally follows predefined paths and known identities.
Security scanners	May create high fan-out and velocity, but often lack result-dependent cross-domain adaptation.
Incident response activity	May be unusually fast and coordinated; use responder identity, case context, and tooling.
Load and resilience testing	May resemble target fan-out or sustained high speed; incorporate change windows and test identifiers.
CI/CD and infrastructure automation	May act across domains; evaluate expected graph, target set, and error-handling behavior.
Worms and fixed malware	Can move at machine speed but may lack adaptive correction. They remain malicious even if PT confidence is lower.

Confounder	Disambiguation Guidance
Shared NAT or proxies	Can merge unrelated actors; rely on richer identity and session relationships.
Telemetry lag or loss	Can create apparent fabricated success or state divergence. Use event-time uncertainty, ingestion health, and corroborating sources.
Eventual consistency	Cloud and distributed systems may expose temporary state mismatches. Apply system-specific settlement windows.
Operator redirection	A human may change goals or scope. Preserve approval and task-transition evidence before labeling drift.
Ambiguous data classification	Sensitivity may depend on aggregation or context. Record provenance and classification rationale rather than relying on public availability alone.

19.1 Adversarial Tempo Shaping

An adaptive adversary may deliberately slow actions, introduce jitter, distribute work across identities or agents, or insert human-like pauses. PHANTOM TEMPO detections should therefore evaluate local bursts, causal adaptation, concurrent branches, and cross-domain continuity rather than relying only on average event rate. Absence of a PHANTOM TEMPO event is not evidence of benign activity or absence of AI assistance.

20. Response Guidance

PHANTOM TEMPO events indicate that defensive decision time may be shorter than ordinary incident workflows. Response **SHOULD** prioritize controls that reduce adversary optionality across domains.

- Revoke or restrict affected identities, tokens, keys, and workload credentials.
- Isolate or constrain the system serving as the operational pivot.
- Block known active paths while avoiding signals that unnecessarily reveal investigative blind spots.
- Preserve high-resolution telemetry, including failures and control feedback.
- Search for sibling paths opened within the same tempo window.
- Review approved automation for changed target sets, commands, credentials, or execution graphs.
- Use human approval for destructive automated response unless predefined safeguards are met.
- Validate agent-reported credentials, sessions, privilege changes, persistence, and exfiltration against authoritative telemetry before acting on the claim.
- Freeze or constrain autonomous execution when repeated state-telemetry divergence or constraint drift reaches a predefined safety threshold.
- Preserve orchestrator state, prompts or task metadata when lawfully available, tool responses, human overrides, and plan revisions for reconstruction.

- Treat a hallucination-correction loop as both an opportunity for detection and a risk of rapid propagation across targets.

20.1 Data Protection and Model Governance

Implementations should minimize collection of sensitive payload content, apply retention and access controls to correlated identity and activity graphs, document model and feature versions, test for biased baselines, and protect assessment datasets against poisoning or unauthorized modification. Automated containment must be bounded by policy, confidence, and reversibility requirements.

21. Governance

PHANTOM TEMPO should be maintained as a versioned public framework. Changes should be proposed through a documented contribution process and should include evidence, rationale, compatibility impact, and test cases.

- Stable identifiers: Published PT-D, PT-B, and PT-C identifiers should not be reassigned.
- Versioning: Major versions may make incompatible structural changes; minor versions may add patterns or fields.
- Evidence requirement: New behavioral patterns should be supported by public research, contributed telemetry, or reproducible testing.
- Vendor neutrality: No vendor-specific implementation should become mandatory unless generalized into an outcome-based requirement.
- Transparency: Material scoring or conformance changes should be documented in release notes.
- Conflict disclosure: Contributors should disclose commercial interests relevant to proposed conformance requirements.
- Integrity-pattern proposals MUST specify the authoritative source used to verify or contradict the asserted state and MUST distinguish missing telemetry from genuine contradiction. Drift-pattern proposals MUST define how legitimate phase changes and operator redirection are excluded.

22. Trademark and Permitted Reference

PHANTOM TEMPO™ identifies the authoritative framework and associated materials maintained by the framework steward. This section is draft brand guidance, not a copyright or trademark license. A separate, counsel-reviewed content license and trademark policy are required before third-party implementation or redistribution terms can be treated as legally operative.

Recommended references:

- “Implements mappings to the PHANTOM TEMPO™ Framework v0.2.”
- “Detects behaviors defined in the PHANTOM TEMPO™ Framework.”
- “Exports PHANTOM TEMPO™ analytic records.”
- “Evaluated against the PHANTOM TEMPO™ assessment methodology.”

Not permitted without written authorization:

- Using PHANTOM TEMPO as the name of an unrelated product, company, or competing framework.
- Claiming certification, validation, endorsement, or partnership that has not been granted.

- Publishing a modified framework as the authoritative PHANTOM TEMPO framework.
- Using the mark in a manner that makes it the generic name for all rapid or AI-enabled attacks.

Before public release, the steward should publish: (1) a copyright license for framework text and data, (2) a separate trademark policy, (3) contribution terms, and (4) a process for resolving disputed mappings or conformance claims.

23. Publication Roadmap

Version	Theme	Planned Deliverables
0.1	Foundation	Core definitions, dimensions, initial pattern catalog, scoring reference, conformance profiles.
0.2	Agentic Integrity and Drift	Operational hallucination definitions, PT-I catalog, three-axis reporting, GTG-1002 mapping, expanded schema and conformance requirements.
0.3	Data, Detection, and Validation	Normative JSON schema, OCSF extension proposal, example SIEM queries, synthetic datasets, integrity and drift test harness.
0.4	Adversarial Validation	Tempo shaping, hallucination residue, goal-drift scenarios, false-positive corpus, red-team assessment suite.
1.0	Stable Release	Stable identifiers, normative conformance specification, maintained knowledge base, formal release process.

24. Initial Research Questions

- Which temporal measurements best separate adaptive agents from fixed automation?
- How should tempo baselines transfer across organizations without exposing sensitive telemetry?
- What minimum event precision is required for reliable adaptation-latency analysis?
- How can cross-domain causal relationships be represented with calibrated uncertainty?
- Which patterns remain robust when adversaries deliberately slow, jitter, or randomize operations?
- How should defenders detect distribution of one operation across many identities or agents?
- What safety constraints are required before automated response is triggered by PT events?
- How can authoritative telemetry be selected and ranked when sources disagree or are eventually consistent?

- Which operational contradictions reliably distinguish model hallucination from ordinary software defects or stale state?
- How can objective drift be measured without requiring access to private prompts or chain-of-thought?
- How should distributed multi-agent systems preserve context and constraints across handoffs?
- Can adversaries intentionally inject false feedback to induce defender-visible hallucination or drift in offensive agents?

Appendix A — Example Analytic Narrative

At 14:02:11.420, workload identity W-17 attempted database authentication and received an access-denied response. Within 2.8 seconds, the same operational cluster queried two separate configuration sources and tested an alternate credential format against a second endpoint. At 14:02:16.901, it retried the original database using a corrected username form and succeeded. During the same 5.5-second interval, a parallel session enumerated schema metadata. The activity was scored PT-C3 because it combined abnormal velocity, parallel hypothesis testing, and a result-dependent correction. Contributing patterns: PT-B1001, PT-B1002, PT-B1005, and PT-B1007. The orchestrator then recorded the credential as validated despite the access-denied event and launched a dependent schema query. The query failed because no authenticated session existed. This added PT-I1001, PT-I1002, and PT-I1004, producing Agentic Integrity AID-3 while Objective Stability remained OS-0.

Appendix B — Example Vendor Claim

Product X implements the PHANTOM TEMPO™ Framework v0.2 at Conformance Level 2. It detects Accelerated Failure Correction (PT-B1001), Parallel Hypothesis Testing (PT-B1002), Compressed Execute-Validate-Adjust Loop (PT-B1005), Unsupported Credential Claim (PT-I1001), and State-Telemetry Divergence (PT-I1004) using identity, cloud audit, endpoint, and orchestrator telemetry. Product X reports tempo and agentic integrity separately and is not certified or endorsed by the PHANTOM TEMPO framework steward.

Appendix C — References

- Sysdig Threat Research Team. “JADEPUFFER: Agentic ransomware for automated database extortion.” Published July 1, 2026.
- MITRE. “MITRE ATT&CK®.” A globally accessible knowledge base of adversary tactics and techniques based on real-world observations.
- MITRE. “D3FEND™.” A knowledge graph of cybersecurity countermeasure techniques.
- Open Cybersecurity Schema Framework. “OCSF Overview.” An implementation-agnostic schema framework for cybersecurity events.
- MITRE ATT&CK. “Legal & Branding” and “Terms of Use.” Reference model for open technical use with controlled trademark usage.
- IETF. RFC 2119, “Key words for use in RFCs to Indicate Requirement Levels,” March 1997.
- IETF. RFC 8174, “Ambiguity of Uppercase vs Lowercase in RFC 2119 Key Words,” May 2017.
- Anthropic. “Disrupting the first reported AI-orchestrated cyber espionage campaign.” Published November 13, 2025; technical report dated November 2025.
- Anthropic. “Mapping AI-enabled cyber threats.” Published June 3, 2026.

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